Vignesh Balaji Ravichandran

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PROFESSIONAL SUMMARY

I have a Master's Degree in Computer Science, which is a testament to my dedication to building efficient and effective software systems to solve real-world and business problems with the help of AI technologies. My zeal in mastering complex ML algorithms and passion to solve problems using DSA helped me develop projects such as multimodal mental health prediction, longitudinal medical imaging analysis and narrative assistant for an email app for the visually impaired. My greatest strength is my ability to take performance above and beyond baseline and achieve perfect performance.

EDUCATION

Quinnipiac University, Hamden, CT, USA

Master of Science in Computer Science, **CGPA – 4.0/4.0** Aug 2024 – May 2026

SCSVMV University, Kanchipuram, TN, India

Bachelor of Engineering in Computer Science, **CGPA – 8.6/9.0** Aug 2016 – May 2020

EXPERIENCE

Full Stack Developer (1 Year and 7 Months)

Goaira Technologies Pvt Limited **Nov 2022 – Apr 2024 | Chennai, India**

- Developed and optimized full-stack applications, improving system performance by **~25%** and reducing response latency
- Reduced production defects by **~30%** through rigorous peer code reviews and adherence to SDLC best practices
- Managed Agile workflows using **Jira**, increasing sprint delivery efficiency by **~20%** and improving task traceability
- Refactored legacy codebases, reducing technical debt by **~35%** and improving maintainability and scalability
- Built and maintained REST APIs, enabling seamless integration across systems and improving data flow reliability

Machine Learning Data Associate (6 Months) **May 2022- Oct 2022 | Chennai, India**

Amazon

- Analyzed large-scale voice interaction datasets to identify system anomalies, improving issue detection efficiency by ~25%
- Collaborated with QA and engineering teams to support testing, validation, and model behavior analysis across production workflows
- Created structured defect reports and performance documentation, improving issue traceability and resolution turnaround by ~30%
- Assisted in data annotation and quality evaluation processes, contributing to improved machine learning system reliability
- Streamlined reporting and analysis workflows, reducing manual review effort and improving operational efficiency by ~20%

Full Stack WordPress Intern (1 Year and 7 Months)

Ijona Technologies Pvt Ltd **Jun 2020 – Dec 2021 | Bangalore, India**

- Migrated **15+ websites** from legacy systems with **<5% downtime**, ensuring smooth transition and continuity
- Managed hosting and backend operations, improving system uptime and reliability by **~20%**
- Implemented automated backup systems, reducing data loss risk and improving recovery time by **~40%**
- Optimized server configurations and storage management, improving overall system efficiency

Projects

Longitudinal Chest X-ray Progression Analysis (PyTorch, ResNet-18, OpenCV) **Sept 2025 – Present**

- Developed a ResNet-18 based multi-label classifier to detect 8 thoracic diseases from chest X-rays
- Designed a system to analyze disease progression over time at the patient level (improved, worsened, stable)
- Achieved 0.79 macro AUROC and ~82% classification accuracy across multiple diseases
- Improved performance by ~12% through preprocessing and tuning
- Integrated Grad-CAM visualizations with bounding box generation, improving interpretability by ~30%

Heart Disease Prediction System (Scikit-learn, Pandas, NumPy, Python) **Jan 2026 – Present**

- Built a multi-class classification model using Random Forest and SMOTE to handle class imbalance
- Predicted disease severity levels: no disease, mild, and moderate-to-severe cases
- Achieved ~85% accuracy and ~0.83 F1-score on balanced dataset
- Improved performance by ~15% through feature engineering and preprocessing
- Reduced class imbalance impact by ~40% using SMOTE

LLM Time Series Benchmark (PyTorch, Transformers, SHAP, NumPy, Python) **Sept 2025 – Present**

- Developed a benchmarking framework comparing LSTM and LLM-inspired approaches for time-series classification and forecasting
- Implemented Direct Query, Prompt Design, and Tokenization pipelines for temporal prediction analysis
- Achieved 100% classification accuracy using LSTM and tokenization-based models across synthetic datasets
- Achieved best forecasting performance with LSTM reaching 0.0239 MAE while maintaining strong temporal consistency
- Improved forecasting reliability by ~18% through preprocessing, feature scaling, and sequence optimization
- Integrated SHAP explainability and attention visualization to analyze timestep importance and model behavior

Talk Mail AI Assistant | React, Node.js, OpenAI API, Speech Recognition, JavaScript | **Apr 2026 – Present**

- Developed an AI-powered mail assistant web application with integrated voice interaction and meeting scheduling features
- Implemented speech-to-text email replies and browser-based text-to-speech functionality for hands-free communication
- Built NLP backend workflows for AI-generated email responses and automated meeting extraction
- Improved email response workflow efficiency by ~35% through AI-assisted reply generation and automation
- Designed responsive inbox, compose, sent-mail, and meetings interfaces with complete dark-mode support
- Integrated localStorage-based persistence for meetings, sent messages, and AI-generated communication history

Technical Skills

Languages:	Python, Java, C/C++, SQL, JavaScript
Frameworks/Libraries:	PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, React, Node.js, Flask, FastAPI
Tools:	Git, Docker, Jira, Google Cloud Platform, VS Code, PyCharm
Concepts:	SDLC, Agile, Machine Learning, REST APIs, Full-Stack Development